Day 15 Problem Set

Geometry Summer Credit | Right-Triangle and Trig Review

Name:	Date:	Period:

Directions

Name the strategy before solving. Show an equation, keep full calculator precision until the end, and include units in context problems.

Part A: Strategy Selection

1.	A right triangle has legs 5 and 12. Name the best strategy and find the hypotenuse.

2.	A 45-45-90 triangle has hypotenuse 12sqrt(2). Name the best strategy and find both legs.

3.	A right triangle has angle 28 degrees and hypotenuse 16. Name the best strategy to find the opposite side.

4.	A right triangle has opposite side 9 and adjacent side 20. Name the best strategy to find the reference angle.

5.	A triangle has sides 11 and 17 with included angle 48 degrees. Name the best strategy to find its area.

Part B: Right-Triangle Skills

6.	Find the missing leg of a right triangle with hypotenuse 25 and other leg 7.

7.	Determine whether side lengths 9, 12, and 15 form a right triangle.

8.	Find the distance between (-2, 1) and (4, 9).

9.	Simplify sqrt(192).

10.	A 30-60-90 triangle has short leg 8. Find the long leg and hypotenuse.

11.	A 45-45-90 triangle has hypotenuse 20. Find each leg exactly.

Day 15 Problem Set

Continued | Trigonometry and Modeling

12.	A right triangle has angle 41 degrees and hypotenuse 22. Find the opposite side.
13.	A right triangle has angle 63 degrees and adjacent side 9. Find the hypotenuse.
14.	A right triangle has opposite side 12 and adjacent side 16. Find the reference angle.
15.	A person stands 45 ft from a building. The angle of elevation to the roof is 39 degrees. Eye level is 5 ft. Find the building height.
16.	From a 75-ft cliff, the angle of depression to a boat is 31 degrees. Find the horizontal distance to the boat.
17.	Find the area of a triangle with sides 13 cm and 18 cm and included angle 52 degrees.
18.	Find the area of a triangle with sides 20 m and 24 m and included angle 35 degrees.
19.	Error analysis: A student uses cosine when the known sides are opposite and adjacent. Explain and correct the choice.
20.	Error analysis: A student uses $A=(1/2)ab \sin(C)$, but C is not the angle between sides a and b. Explain the problem.
21.	Challenge: A triangular solar panel has sides 16 ft and 21 ft with included angle 72 degrees. Find the area and explain what the units mean.